

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	VIMALA K	Reg. No.:	192424276
Date:	6/8/26	Duration:	60 minutes

Code of Conduct

I VIMALA K (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Vimalak

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description
 - Scope of the System

- Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
- List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
- Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

(15%)	sections.			components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Intelligent Electric Vehicle Charging Network Management System UML DIAGRAM

CSA1014 -- Software Engineering

Student Name: VIMALA K

Register Number: 192424276

Department: B.Tech AI&DS

Scenario Analysis

Business Domain

- Smart Transportation
- Electric Vehicle Infrastructure
- Energy Management
- IoT-based Charging Network

Existing Challenges

- Long waiting time
- Poor charger availability information
- Uneven charger usage
- Lack of real-time monitoring

Objectives

- Monitor charging stations
- Predict waiting time
- Optimize charger allocation
- Generate energy reports
- Improve customer experience

Functional & Non-Functional Requirements

Functional Requirements

- User Registration
- User Login
- Search Charging Station
- View Charger Availability
- Book Charging Slot
- Start Charging Session
- Stop Charging Session
- Make Payment
- Receive Notifications
- Generate Energy Reports
- Manage Users
- Manage Charging Stations

Non-Functional Requirements

Requirement	Description
Performance	Fast response time
Security	Secure login and payment
Reliability	Continuous system operation
Availability	24x7 service
Scalability	Supports increasing users
Maintainability	Easy updates and maintenance
Usability	User-friendly interface

Actors & Use Cases

Primary Actors

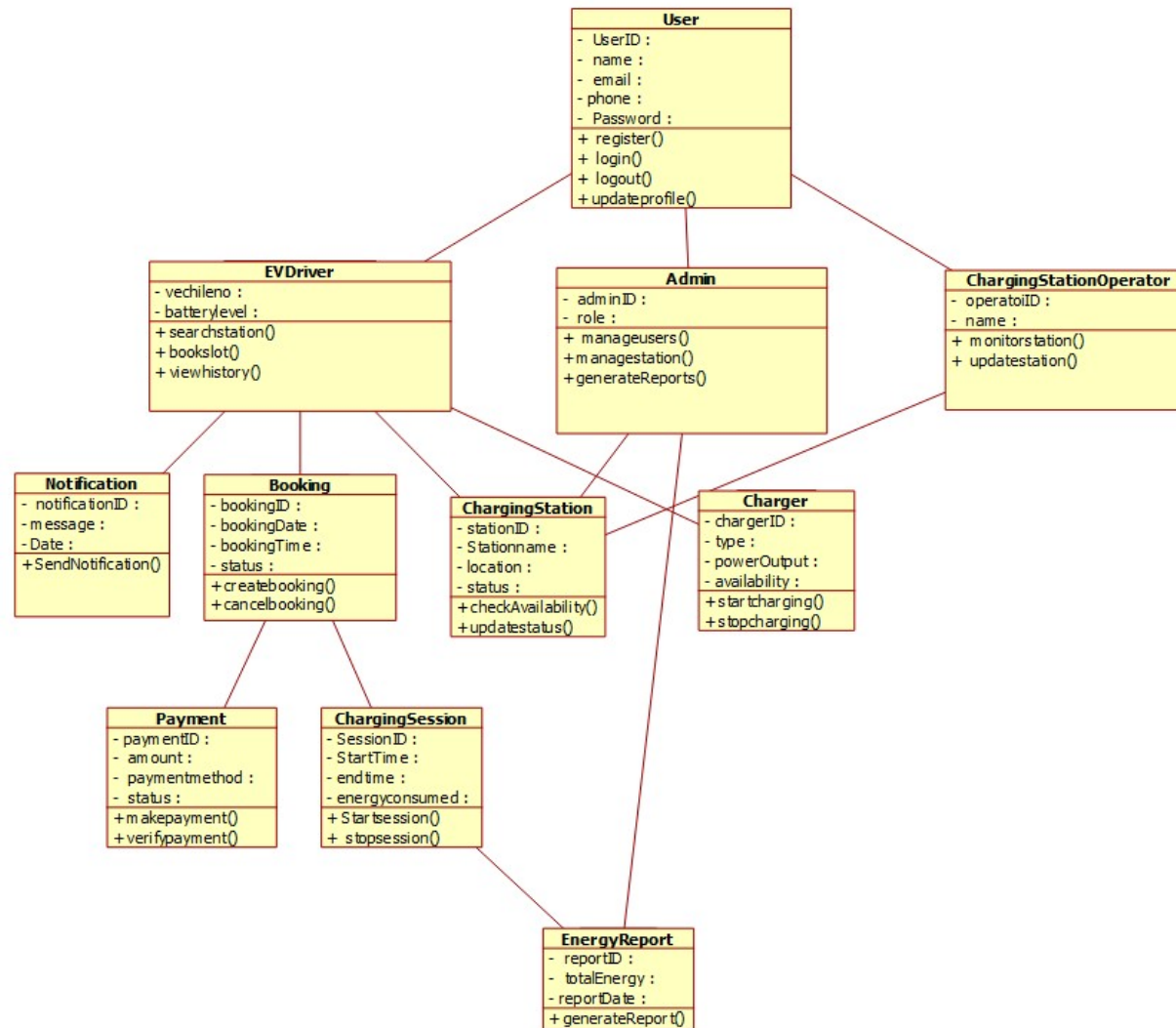
- EV Driver
- Charging Station Operator
- Administrator

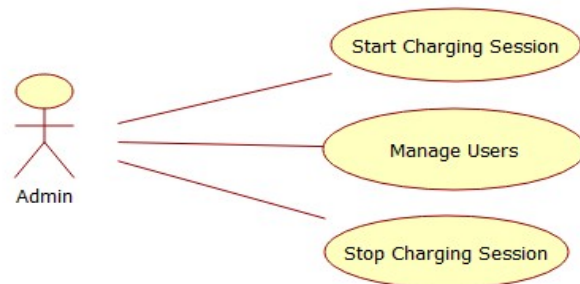
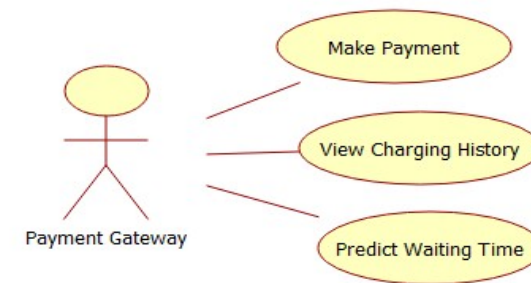
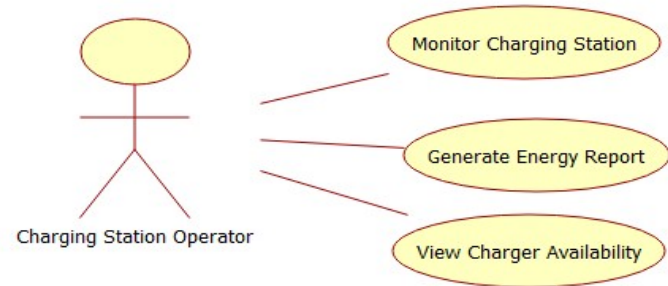
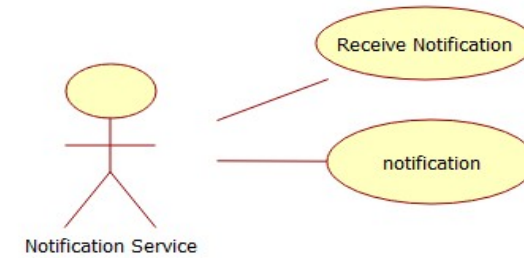
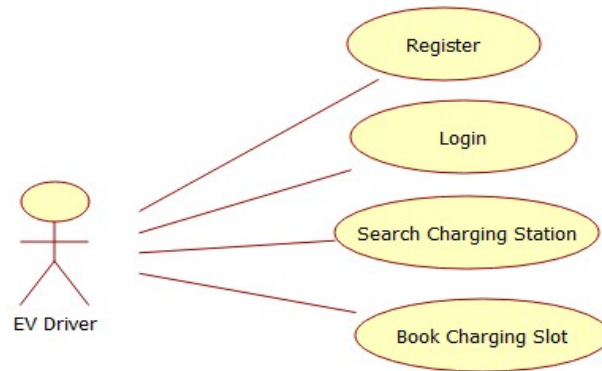
Secondary Actors

- Payment Gateway
- Notification Service

Major Use Cases

- Register / Login
- Search Charging Station
- View Availability
- Book Charging Slot
- Start Charging
- Stop Charging
- Make Payment
- Receive Notification
- Generate Reports
- Manage Users
- Manage Charging Stations





Software Requirement Specification (SRS)

SRS Sections

- Introduction
- Problem Description
- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Description
- Data Requirements
- External Interface Requirements
- System Features



Data Requirements & External Interfaces

Data Requirements

- User
- Charging Station
- Booking
- Charging Session
- Payment
- Notification
- Energy Report

User Interface

- Mobile Application
- Web Dashboard

Hardware Interface

- IoT Charging Stations
- GPS

Software Interface

- Payment Gateway
- Notification API
- Google Maps API

Communication Interface

- Internet
- Wi-Fi
- 4G / 5G

Assumptions, Constraints & Requirement Validation

Assumptions

- Stable Internet connection
- IoT-enabled charging stations
- Registered users
- Cloud infrastructure available

Constraints

- Technical
- Operational
- Economic
- Legal
- Time

Requirement Validation

- ☒ Stakeholders identified
- ☒ Functional requirements completed
- ☒ Non-functional requirements completed
- ☒ Use Case Diagram prepared
- ☒ Class Diagram prepared
- ☒ Activity Diagram prepared
- ☒ SRS is complete and consistent

Conclusion

The system provides:

- Real-time charging station monitoring
- Smart charger allocation
- Waiting time prediction
- Online slot booking
- Secure payment
- Energy report generation
- Better customer experience

Future Enhancements

- AI-based demand prediction
- Smart pricing
- Renewable energy integration
- Predictive maintenance
- Vehicle-to-Grid (V2G)

Thank You